

José Miguel Ramírez Gutiérrez

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EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

B.S. in Robotics and Digital Systems Engineering (GPA: 3.7/4.0)

Querétaro, Mexico

Aug. 2024 – Expected 2028

- Socioeconomic Scholarship

EXPERIENCE

Cloud Integration & IT Specialist

binarysails

2026 – Present

Querétaro, Mexico

- Verified and standardized Microsoft OneDrive functionality, remote sharing, and synchronization across multiple external computers, ensuring consistent file access; provided remote troubleshooting and testing via RustDesk.
- Authored an official instructional syllabus documenting cloud collaboration techniques and synchronization states; delivered training for transition from local storage to OneDrive.
- Standardized organization of electronic and firmware files within shared drives, improving traceability and delivery of project documentation.

Automation Developer (PEPS Validation)

Valeo (Contractor)

June 2026

Querétaro, Mexico

- Automated manual Key Fob functional validation by developing a complete hardware/software positioning system, saving engineers up to 8 hours of testing daily.
- Designed and validated an autonomous workspace calibration system using 4 HC-SR04 ultrasonic sensors with a halving-approach algorithm, achieving ± 1 cm positioning accuracy over a 390×1300 mm test area; diagnosed and resolved sensor crosstalk through circuit-level analysis.
- Built a Python/Tkinter GUI for real-time lab instrumentation control and automated execution of validation scripts, centralizing hardware communication.
- Documented the system, calibration parameters, and test procedures for handover to validation engineers.

PROJECTS

Custom Linux Single Board Computer (SBC) | *KiCad, Allwinner V3s, Hardware Design*

July 2026

- Engineered a custom 4-layer PCB around the Allwinner V3s (ARM Cortex-A7) with EA3036CQB PMIC, Ethernet PHY, and RTC; differential-pair routing for USB 2.0 High-Speed (480 Mbps) with controlled impedance.
- Optimized layout for automated SMT manufacturing (PCBA), resolving centroid rotation and footprint constraints through DRC/ERC review.

ESP32 AI Voice Assistant | *MicroPython, KiCad*

May 2026

- Designed custom PCB with ESP32, INMP441 I2S microphone, PAM8403 amplifier, and SH1106 OLED; generated production-ready Gerber files and integrated Groq API (Whisper + LLaMA 3.1) via Flask backend on Render with VAD wake-word detection.

DetectaGas – IoT Gas Leak Prevention System | *ESP32, Rust, React, MQTT*

December 2025

- Engineered an embedded IoT device using an ESP32 microcontroller and MQ-5 gas sensor circuit to rapidly reduce response time to hazardous LP gas leaks.
- Built a concurrent Rust/Axum backend and React web interface, processing continuous sensor telemetry via an MQTT broker.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, C#, JavaScript, MATLAB, VHDL

Embedded & Hardware: ESP32, ESP8266, Arduino, ATmega328P, KiCad, PCB & Schematic Design, Circuit Analysis, Datasheet Interpretation

Communication Protocols: I2S, SPI, UART, MQTT

Cloud, Remote & Office: Office 365/OneDrive (verification, sync, remote sharing), RustDesk (remote diagnostics/testing), documentation standardization, user training

Networking: Cisco Packet Tracer, switching/routing fundamentals

Software & Tools: MicroPython, Arduino IDE, Axum, MySQL, Git, React, Flask

Spoken Languages: Spanish (Native), English (Intermediate – reading manuals and technical correspondence)